

# SlackSmith

## A GenAI RTL optimization loop that proves each RTL change correct, then measures whether it helps

Nebula @ BITS Goa 2026, Track A (Digital): Constraint Optimization through RTL Enhancement Using Generative AI

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Repository: [github.com/nilaymastaadmi/nebula-slacksmith](https://github.com/nilaymastaadmi/nebula-slacksmith). Demo video: `slacksmith_demo.mp4` in the repository root. Every number in this report was produced by running the flow; the command that produced each one is in the cited experiment directory.

### 1. Summary

An LLM proposes RTL transforms against a timing report; SlackSmith applies one only once it is **proven correct** and keeps it only if timing **measurably improves**. What is new is the checking: a fingerprint that catches a constraint edit (Go), proof obligations generated from the declared transform type, a null control that catches miter artifacts, and a sealed exam that grades the checker itself (SlackBench). Measured with those, the LLM fails in two ways: transforms that are **wrong**, and transforms that are **correct and aimed at the wrong variable**. On a 55K-cell benchmark and two external designs, **no proven RTL transform bought a timing gain that survived the physical flow**.

**Failure one, correctness.** We asked an LLM for transforms against real timing reports, froze every proposal before running any check, and ran two pre-registered batches. **3 of 12 were formally refuted, 1 more was rejected at precondition, 1 is unresolved.** Of those three, **two are corroborated by a control that closes and one is not:** P5's control does not close at any budget tried, though the one artifact class found in our own miter is ruled out for it (§7.6). One of the three parses, elaborates, passes every precondition, cuts 208 cells, and is wrong. The design's own shipped firmware misses it; 20,000 random instruction vectors miss it; the formal gate caught it in **46 seconds** with a concrete counterexample. **The gates that agentic RTL tools actually ship with would have accepted a broken rewrite.** That count started at 4 refuted; it is 3 because batch 2 caught **our own gate manufacturing a refutation** (§7.1, §9).

**Failure two, relevance.** Then we asked whether the transforms that *are* correct actually help. Mostly they did not, for a measurable reason: the binding paths were **59% to 91% fanout-attributable delay**, which is a net's load, and physical buffering is built to fix that. **OpenROAD repair\_design, changing zero lines of RTL, closed all three groups at 20.2% area.**

**We over-claimed that and a later measurement caught us.** Earlier drafts said *no RTL rewrite shortens a net's load delay*. The absolute is false: on a path scored **91.4% fanout-attributable**, an FSM re-encoding bought **+3.185 ns** and an unattended fanout split **+1.414 ns**, no group paying (§7.6, §7.7). The honest claim is a **ratio that must name its pair**. Like for like, SDC v3, zero-parasitic, `clk_b`: the FSM re-encoding buys **+3.185 ns** while the physical lever takes the same group **-18.957 to +5.6** in the same run. **About one eighth.** Other pairs give a third to a seventeenth by mixing SDC versions and parasitic regimes, so one pair under one model is quoted. **The quantity that matters most is now measured (§7.8).** Composed into one file, the three proven transforms are worth **+5.165 ns** on `clk_b` before wires, 54% of the sum of their parts; after the ABC buffering lever they are worth **0.000**; after `repair_design` **-0.237 ns against a 0.24 ns perturbation floor** measured on five netlists. No RTL timing gain in this report survives the physical lever on the paths it was aimed at, and the classifier had routed every one of those paths to that lever before any of it was measured.

**One limit that belongs here and not in a footnote. Under a correct classifier the router never selects the RTL lever on this benchmark at all;** every in-loop RTL result used `--force-lever rtl`, logged as `lever_forced`. That is a finding about the benchmark. The unattended backend does now run, **N = 1** (§7.7). On the two external designs where the router chose RTL unforced, four proven transforms bought **0.000, 0.000, -0.268 and -0.421 ns** (§7.8).

So SlackSmith routes twice: the **proof obligation** by declared transform type, and the **fix** by measured path pathology, which the measurement forced on us. The two levers are sequential rather than alternative, and where they run out we say so.

#### 1.1 What the agent actually automates, and what we cannot claim about it

The practical case for this tool is not that it optimizes better than an engineer. It is that the loop below runs as **one command**, and every step of it is otherwise a human reading a report and deciding:

Fingerprint the SDC (Go); synthesize and time; **classify the binding path** by fanout against depth (~2 s), routing *away* from RTL when the lever is physical; propose (5 to 9 min unattended); **generate and discharge the proof obligation** from the declared type (1 s to 222 s); re-time; accept or revert. End to end, the v2 benchmark closes in 2 iterations in a median **112.7 s** on one core (five protocol runs, 83.9 to 152.7 s; outside the protocol the same command has measured 47 to 833 s on this machine; `experiments/loop_runtime/`); the unattended run took **456 s**, one sample.

**What we cannot claim: a speed-up ratio.** No engineer was timed doing the same work, so there is no denominator. What is counted instead is the manual work replaced: the obligation generator writes **326 lines** (177 of code, 46 port

connections, 10 properties) across three branches (experiments/speedup\_step/). What is measured is the automation's own cost and the decisions it takes unaided: it reverted three formally proven transforms that made timing worse and declined to spend a proposal on a 91.4% fanout path, both unasked. Synthesis dominates the time; the proposer is minutes, the proof seconds, so a faster proposer would not help.

## 2. The problem, and what is actually new here

**Three things here are ours, stated before the concessions that narrow the rest.** **SlackBench grades a verification methodology**, not a design and not a testbench (§7.4): eight transform pairs with ground truth committed before any checker ran, our own gate scored among them. **Go catches the one attack no equivalence checker can see** (§5.2): on a byte-identical netlist, changing only the constraint file closes a group, and every checker we own calls the two designs equivalent, correctly, because they are the same file. **A null control runs before any refutation is reported** (§9), because our own gate manufactured three refutations that were harness artifacts.

Timing closure is manual because static timing analysis speaks in cells and nets while RTL speaks in always blocks, and an LLM bridges those well. The difficulty is not proposing a rewrite but knowing whether it is correct. Insert a pipeline stage and the design is equivalent only under a latency offset the checker must be told about, so the profitable transforms are forbidden because the available checkers cannot express them.

**Scoped precisely, because the loose version is false.** Generation-mode agents write any latency they like. The accurate claim is that we found no published system that both *changes* latency and *discharges a formal obligation for it*. Dr. RTL (ICCAD 2026) preserves micro-architecture "including pipeline latency"; RTLScout gates on a Verilator testbench with abc cec secondary, and CEC structurally cannot see an added register.

**Four neighbours narrow the claim, two of them badly.** **ASPEN** (MLCAD 2025) and **ROVER** (TCAD 2024) pair rewriting with combinational obligations, and ROVER's propose-then-EC-gate loop is **structurally our shape with the search replaced by a model**. **EquivFusion** (arXiv 2604.16571) already derives the obligation from a declared scope; our taxonomy is finer, five types against two, and that is honestly the whole difference. **ElasticMiter** (ASPLOS 2025) proves in Coq what our elastic branch checks, so "agreement modulo k cycles under back-pressure" is **not ours**. What remains: five obligation types routed *automatically from a declared transform type* on an open-source stack, and Dr. RTL's 14% sequential-equivalence failure rate is what happens when that routing is left implicit.

**Measuring how often the model is wrong is not new either.** Dr. RTL reports an 86% SEC pass rate and **RealBench** (arXiv 2507.16200) reports **44.2% of GPT-4-Turbo Verilog that passes RTLLMv2's testbenches failing formal verification**. Both measure wrong code escaping a *weak* oracle. What we found measured nowhere is what escapes a checker of the **wrong type** (§6.2, §7, experiments/slackbench/).

**What we built.** Transforms are *typed*: the model emits a declared transform type rather than free-text Verilog, and that declaration mechanically determines which obligation is generated. One that cannot be discharged is never reported as a result. The interface classifier (§6.1) decides rigid versus elastic and the declared latency delta decides padded versus plain, so none of the routing is left to the model.

| declared                 | obligation branch                      | discharged by                |
|--------------------------|----------------------------------------|------------------------------|
| k = 0, state-preserving  | combinational / sequential equivalence | EQY, yosys-abc dsec          |
| k > 0, rigid interface   | k-padded miter                         | SymbiYosys, BMC + PDR        |
| k > 0, elastic interface | stream equivalence                     | SymbiYosys + cover           |
| k = 0, re-encoded state  | mapped-state equivalence               | SymbiYosys, sequential miter |
| k = 0, register moved    | retiming                               | SymbiYosys, sequential miter |

All five are discharged on blocks of the benchmark itself. Two pieces of §6 evidence are **fixtures by construction and labelled as such**: the classifier table in §6.1 and the four mutants in §6.2. Those test the *classifier* and the *checkers*, which is what they are for.

## 3. Deliverable coverage

| #   | Organizer deliverable                                                   | Where it is evidenced                                                                                                      |
|-----|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 1   | RTL timing analysis framework                                           | §5, sdc/bench_top.sdc, tools/remesure.py                                                                                   |
| 2   | GenAI-based RTL optimization engine                                     | <b>Partial.</b> §7, experiments/llm_proposer/, tools/gate_proposal.py                                                      |
| 3   | Critical path and timing violation analysis                             | §5, docs/measurement-methodology.md                                                                                        |
| 4   | Optimized RTL implementation                                            | <b>Partial.</b> §7.8, experiments/composed_rtl/aes_key_mem_composed.v (composed, proven), plus rtl/rv32i_core_P{1,2,3,6}.v |
| 5   | Timing, frequency and PPA comparison                                    | §8, experiments/ppa/                                                                                                       |
| 6   | Formal equivalence verification report                                  | §6, §7.5, experiments/*/NOTES.md + logs                                                                                    |
| 7   | Interactive demo                                                        | <b>demo/explorer.html</b> , §10                                                                                            |
| n/a | Benchmark: 5 async domains, generated clocks, CDC, dividers, ~50K cells | §4, rtl/bench_top.v                                                                                                        |

**The four optimization classes the objectives name, each scored separately.** A class is covered when the engine *proposes* it and the gate *routes* it, not when the report mentions it.

**Provenance is stated per proposal, because the three tiers are not equivalent evidence. Frozen:** written against a timing report and committed before any check ran. **Handoff:** written against live loop state, but through the session driving this project, which carries full context. **Unattended:** claude -p, no human, no context beyond the prompt.

| class               | proposals                           | provenance | branch | honest status                                                                                                                                   |
|---------------------|-------------------------------------|------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| logic restructuring | P1, P2, P3, P6, P4, A1, A4, A5      | frozen     | 1      | 8 of the 12 frozen proposals                                                                                                                    |
|                     | online_proposer O1, O2              | handoff    | 1      | O1 kept, O2 proven and 11.434 ns worse                                                                                                          |
|                     | fanout_replication_round_key_update | unattended | 1      | <b>PROVEN, +1.414 ns on every group</b> (§7.7)                                                                                                  |
| pipelining          | P5, A3                              | frozen     | 2      | <b>proposed twice, proven never.</b> A3 refuted, P5 refuted-uncorroborated. The proven pipeline_cut_rigid in §6 is hand-built                   |
| retiming            | missing_classes O1                  | handoff    | 5      | PROVEN, costs 4.616 ns (§7.6)                                                                                                                   |
| FSM optimization    | missing_classes O2                  | handoff    | 4      | PROVEN, <b>+3.185 ns</b> , the largest RTL gain on that group (§7.6). An unattended proposal on an external design also reached branch 4 (§7.2) |

**The engine recommends four of four classes and the evidence is not equal.** Logic restructuring is demonstrated at all three tiers including unattended. Pipelining is proposed at the frozen tier and never proven. **Retiming and FSM optimization exist only at the handoff tier**, so a model with this project's context wrote them, not a blind one: before 11 Sept the gate rejected both by construction (§7.6), so no frozen batch could have held one.

Two of the four were unreachable by construction and why was established only on 11 Sept (§7.6). dretime also appears inside the ABC physical-lever script, which is a mapping-level pass and **not** the engine proposing a retiming; it is not counted here. **Two experiments each number their proposals O1 and O2**, experiments/online\_proposer/ and experiments/missing\_classes/; they are always named with their directory here, and the collision is flagged rather than fixed by renumbering a registered artifact after its results existed.

## 4. The benchmark

bench\_top is **55,413 standard cells** hierarchical, the figure this report uses throughout; tools/bench\_size.py and the demo measure **48,616** on the later flattened, buffered-and-sized netlist, because flattening collapses redundant decode across the module boundary (§7.3) and saves more than buffering adds. Both are ~50K, both are cross-checked against a fresh Yosys flatten, and the pair is stated here so a viewer who sees one number and a reader who sees the other are not looking at a contradiction. It has five independent asynchronous clock domains, each with its own active-low async reset and each driving at least one in-RTL generated clock.

| domain | clock | divider | contents                                                               |
|--------|-------|---------|------------------------------------------------------------------------|
| A      | clk_a | /2      | 8-bit MAC datapath + <b>RV32I core</b> running a real instruction loop |
| B      | clk_b | /3      | 10-state control FSM + LFSR + <b>AES-128</b>                           |
| C      | clk_c | /4      | UART-style shift block                                                 |
| D      | clk_d | /5      | reloading timer, two compares                                          |
| E      | clk_e | /2      | 16 x 32-bit config file + <b>AES-128</b>                               |

Crossings form a ring A→B→C→D→E→A: gray-pointer async FIFOs on every multi-bit crossing (16, 8 and 32 bits wide), two-flop synchronizers carrying toggles on every single-bit one. **Four of the five crossings both launch and capture on generated clocks.**

**The /3 and /5 dividers are the deliberate difficulty.** An odd ratio needs two counters, one per edge, ANDed. Simulated with reset released off the source grid at 13.3 ns, /2 /3 /4 /5 measure **exactly 10.000, 15.000, 20.000 and 25.000 ns at 50.0% duty** (experiments/clkdiv\_sim/), and the SDC must describe generated clocks whose edges derive from both source edges.

Third-party content: the AES-128 core is secworks/aes, BSD-2-Clause, vendored unmodified under rtl/aes/ with its license and a THIRD\_PARTY.md. The RV32I core is ours, verified against a golden C++ instruction-set simulator over a 400-seed differential regression **whose evidence lives in the rv32-dsp-soc repository, not this one.**

## 5. Timing analysis framework, three findings, and the gate below all of them

The SDC is **written once and frozen** before any optimization runs. The agent never edits constraints, and timing exceptions including multi-cycle paths are excluded from the transform set entirely, so no reported improvement can come from relaxing the measurement.

Generated-clock -edges for the odd dividers were derived from the divider's edge arithmetic (rise @ 2\*DIV, fall @ 2\*(DIV + (DIV+1)/2) - 1, next rise @ 4\*DIV, so DIV=3 gives {6 9 12}) and cross-checked three independent ways: the hand trace, an Icarus simulation, and OpenSTA's report\_clock\_properties. All three agree to the decimal.

One `set_clock_groups -asynchronous` over the five domains is also what exempts every synchronizer path from ordinary setup/hold analysis. No per-path exception is used anywhere except a false path from the five asynchronous reset ports, which Go counts, and **no `set_multicycle_path` appears in the file at all**, because a multicycle exception manufactures slack without changing the design.

Three findings changed how we report every number (`docs/measurement-methodology.md`):

**A single library cell was worth 4.83 ns of pure artifact.** The first critical path put 19.5 of 33.0 ns in *two cells*: `sky130_fd_sc_hd__lpflow_isobufsrc_1`, a low-power isolation cell ABC selected on area cost. Excluding the `lpflow` and probe families, as standard `sky130` flows do, moved WNS **-27.37 to -22.54 with zero RTL change**.

**The measurement tool has a zero noise floor:** swapping a module for *itself* returns 0.000 delta on every path group. **But local RTL changes have non-local effects.** Given that null control, the +1.398 ns a `domain_b`-only change produces on `clk_a` is not noise, it is ABC's global mapping moving an unrelated group. **Reporting rule adopted:** a transform's effect is the delta in the group it touches, and movement elsewhere is reported separately, never folded into the claimed benefit.

## 5.1 Setting a target that means something, and a fix that did nothing

The v1 periods were illustrative, chosen when the benchmark was 3,584 cells; at 55,413 an 8 ns `clk_a` target demands four times what a single-cycle RV32I with `async-read` memory reaches in `sky130`. So we measured what each domain requires and set `sdcbench_top_v2.sdc` about 10% tighter, keeping v1 unchanged as the frozen record: revising a target with disclosure is not editing constraints mid-campaign, which stays forbidden. `sdcbench_top_v3.sdc` (§7.3) applies the same method to the buffered flow. Under v2 the baseline **meets** `clk_a` at +1.333 ns and the two AES-bound domains sit at -4.957; **read those against §8**, which shows what they become once wires exist.

**Finding 1 above was correct and not in effect for eight commits:** the exclusion regex expected unquoted cell names, the liberty quotes them, and the flag string came back empty, so every netlist in between carried 203 `lpflow` cells and the 12.8 ns artifact. Found by reading a path report; there was no test. All affected numbers were re-measured, the `clk_a` baseline carried **4.62 ns** of artifact, deltas moved 40 to 50%, and **no qualitative conclusion changed**. The exclusion is not uniformly beneficial: `clk_a` gains 4.62 ns while `clk_b` and `clk_e` each lose 1.97.

## 5.2 Go, constraint integrity: the one attack no equivalence checker can see

Everything above trusts the SDC. We measured what that trust is worth. On **one netlist, 26,958 cells, byte-identical in every row**, with no RTL edit, no resynthesis and no gate resized, the only thing varied was the constraint file:

| appended constraint                                              | clk_e                  |
|------------------------------------------------------------------|------------------------|
| none (honest baseline)                                           | <b>-0.319 VIOLATED</b> |
| <code>set_multicycle_path 2 -setup -to &lt;endpoint&gt;/D</code> | -0.295                 |
| <code>set_false_path -to &lt;endpoint&gt;/D</code>               | -0.295                 |
| <code>set_multicycle_path 2 -setup -from clk_e -to clk_e</code>  | <b>+4.860 MET</b>      |

One line closes the group, worth **+5.179 ns**, and it changes nothing at all. Earlier drafts added "more than our best proven RTL transform (+4.925)"; that comparison is **withdrawn**, because the +5.179 is measured on a 26,958-cell flat netlist in one domain and the +4.925 on the 55,413-cell hierarchical benchmark. Different fixtures, so the sentence was doing rhetorical work the measurement does not support. The mechanism needs no comparison to land. **Every checker we own returns "equivalent" on that pair, correctly, because the two designs are the same file.** A project whose entire correctness story is functional equivalence has no defence against a constraint edit.

The narrow version does not pay: aiming the exception at the reported endpoint buys 0.024 ns because the worst path moves to the next endpoint. Only the domain-wide exception works, and that is a conspicuous line in an SDC diff, provided anyone looks.

So the policy became a gate. **Go runs before G1:** SHA-256 the SDC actually loaded, compare it against the registered digest, count the timing exceptions, and refuse to report any measurement taken under constraints that differ from the frozen file (`sdcbenchmark_fingerprint()` in `tools/slacksmith.py`, `--expect-sdc-sha`). It is cheap and it closes the one surface G1 to G5 structurally cannot reach.

`experiments/sdc_integrity/` is **exploratory, not pre-registered**, and says so in its own notes: it demonstrates a mechanism rather than testing a hypothesis.

## 6. Formal equivalence: five branches, discharged on benchmark blocks

| transform                                              | branch                | verdict                 | engine      |
|--------------------------------------------------------|-----------------------|-------------------------|-------------|
| <code>fsm_reencode(domain_b)</code>                    | 4, mapped-state       | <b>PROVEN unbounded</b> | PDR         |
| <code>mux_priority_to_parallel(domain_b_onehot)</code> | 1, scoped             | <b>PROVEN unbounded</b> | PDR         |
| <code>pipeline_cut_rigid(domain_a)</code>              | 2, k-padded           | <b>PROVEN unbounded</b> | PDR         |
| <code>pipeline_cut_elastic(sync_fifo)</code>           | 3, stream equivalence | <b>PROVEN unbounded</b> | PDR + cover |
| one-hot state invariant                                | supporting lemma      | <b>PROVEN unbounded</b> | k-induction |

**We measured what the existing checkers do**, rather than asserting it. Given a correct latency-changing transform, with a positive control run first in every case:

| checker        | control    | correct latency+1 transform                                   |
|----------------|------------|---------------------------------------------------------------|
| yosys-abc cec  | equivalent | <b>cannot build the miter</b> ("different number of latches") |
| yosys-abc dsec | equivalent | <b>NOT EQUIVALENT</b>                                         |
| EQY            | PASS       | <b>FAIL</b> , 1/1 partitions                                  |
| k-padded miter | n/a        | <b>PASSED, unbounded</b>                                      |

cec cannot express the question; dsec and EQY express it and correctly answer no, because the designs are not cycle-for-cycle equivalent. None can express equivalence *modulo k cycles*, so a pipeline gated on any of them can only ever reject a latency change. Measured evidence for the routing decision, not an argument.

**We mutation-tested our own checker, and it failed.** A mutant reproducing a documented deadlock, confirmed by simulation to never produce output, was **PROVEN by PDR in 0 seconds**: the stream-equivalence assert's guard was unreachable and the property vacuously true. Fixed with a cover that distinguishes the real design (REACHED) from the mutant (UNREACHABLE).

## 6.1 Choosing the branch: the interface classifier

A k-padded obligation is *wrong* for an elastic interface: pointed at a valid/ready pair with back-pressure it is **refuted in 0 seconds on a correct design**, because no fixed cycle offset exists. **A tool that emits the wrong obligation reports a correct transform as broken.** So the branch is chosen in three passes, each able to overrule the last: **lexical** (handshake ports by name), **structural** (does the candidate reach a flop's D or enable cone), and **formal** (prove output stability under back-pressure).

| module                   | lexical       | structural      | formal        | verdict      |
|--------------------------|---------------|-----------------|---------------|--------------|
| mac_ref (rigid)          | no candidate  | n/a             | n/a           | RIGID        |
| mac_vr_ref               | out_ready     | reaches \$dff.D | PASSED        | ELASTIC      |
| alias_names (vld/rdy)    | o_rdy         | reaches \$dff.D | PASSED        | ELASTIC      |
| axi_style (AXI prefixes) | m_axis_tready | reaches \$dff.D | PASSED        | ELASTIC      |
| costume_ready            | out_ready     | <b>REJECTED</b> | <b>FAILED</b> | <b>RIGID</b> |

costume\_ready earns the machinery: handshake-shaped names, not an elastic interface, rejected by two independent arguments. Limits: pass 3 is bounded at depth 16, and flow control that misses the lexical pass is classified rigid, the unsafe direction.

## 6.2 Why simulation is not a substitute, measured on four mutants

**Before the LLM experiment, four hand-built mutants of a known-correct transform, with a control every method passes.** Two of the three invalid mutants escaped a realistic simulation gate: the classic pipelining bug (stage 2 adding the current operand to a product one cycle old) and one wrong on roughly one input in a million that survived 20,000 random vectors; formal refuted both instantly. §7 reproduces this on a real proposal; §7.4 turns it into a suite.

## 7. The GenAI engine, and the experiment we pre-registered

The model receives the OpenSTA critical-path report, the target RTL and the typed transform schema, and emits a declared transform type plus replacement source. The declaration selects the obligation; tools/gate\_proposal.py runs five gates in order: **parse, elaborate, precondition, formal, timing**.

To measure this honestly we **pre-registered the experiment before writing any proposer code**, and git proves the ordering (04fa59c precedes 115fc03 precedes the results). The registration fixed N = 6, fixed the gates, and fixed the anti-tuning rule: *all six proposals committed before any gate ran, none editable afterwards, all six reported regardless of outcome*.

|    | transform             | declared | parse | elab | precond | formal         | timing (clk_a) |
|----|-----------------------|----------|-------|------|---------|----------------|----------------|
| P1 | addsub sharing        | k=0      | ✓     | ✓    | ✓       | PROVEN         | -1.555         |
| P2 | shifter sharing       | k=0      | ✓     | ✓    | ✓       | PROVEN         | <b>+0.485</b>  |
| P3 | comparator sharing    | k=0      | ✓     | ✓    | ✓       | PROVEN         | -2.102         |
| P4 | mux priority→parallel | k=0      | ✓     | ✓    | ✓       | <b>REFUTED</b> | n/a            |
| P5 | pipeline cut          | k=1      | ✓     | ✓    | ✓       | <b>REFUTED</b> | n/a            |
| P6 | branch cmp sharing    | k=0      | ✓     | ✓    | ✓       | PROVEN         | -1.615         |

**4 proven, 2 refuted, 1 improved timing.** The registered primary bar was met by P2, which is also the only proposal both smaller (-351 cells) and faster.

### P4: the proposal that was wrong in the way that matters

P4 one-hot decoded funct3 and OR-ed the masked arms. It parses, elaborates, passes preconditions and saves 208 cells. EQY refuted it in 46 seconds with the witness a = ae19f605, shamt = 7, isolating **one failing partition out of 447**: alu\_out.

It had folded the shift arms into a ternary:

```
wire [31:0] r_shr = alt ? ($signed(a) >>> shamt) : (a >> shamt);
```

A conditional operator's signedness derives from *both* branches. Pairing a signed branch with an unsigned one makes the whole expression unsigned, that context propagates back into the operands, and >>> silently degrades to a logical shift. SRA is then wrong for every negative operand.

rv32i\_core.v carries a six-line comment warning about exactly this, **ten lines above the code being transformed**. The model had that file as input and made the documented mistake anyway.

## The four-checker column on P4

EQY (formal) **CAUGHT** it in 46 s. Simulation on the design's own shipped firmware loop, 400 cycles: **MISSED**. Simulation on 20,000 random instruction words: **MISSED**. A directed SRAI on a negative operand: caught.

We ran the directed probe specifically so the two misses could not be mistaken for a phantom defect: the bug is fully visible to simulation (gold ffffffff, gate 0fffffff). Both misses are stimulus weakness. The shipped firmware loop contains no shift instruction at all, so it can never reach the bug. **The verdict tracks stimulus quality, not bug severity.**

P5 shows the k-padded obligation doing its job in the other direction: refuted in 1 second because alu\_out feeds the register file and PC, so a declared k = 1 shift is not what the transform actually does.

## 7.1 Batch 2, registered separately, on the AES key memory

Batch 1's registration required any second batch to be registered separately with its N added to the trial count (dbcd7d3 precedes 004695c precedes every result). **Trial count: 16**. Batch 2 targets aes\_key\_mem, which holds the worst path on clk\_b and clk\_e; u\_aes\_b and u\_aes\_e are two instances of one module, so every edit is measured twice.

|    | transform              | declared | precond  | formal     | clk_b         | clk_e         |
|----|------------------------|----------|----------|------------|---------------|---------------|
| A1 | mux bank split         | k=0      | ✓        | PROVEN     | +2.020        | +2.020        |
| A2 | decode duplication     | k=0      | ✓        | UNRESOLVED | n/a           | n/a           |
| A3 | read port register     | k=1      | ✓        | REFUTED    | n/a           | n/a           |
| A4 | one-hot read select    | k=0      | ✓        | PROVEN     | <b>+4.925</b> | <b>+4.925</b> |
| A5 | reset unroll (control) | k=0      | ✓        | PROVEN     | +0.436        | +0.436        |
| A6 | key mem parity split   | k=0      | REJECTED | n/a        | n/a           | n/a           |

**A4 takes clk\_b from -4.957 to -0.032**, zero-parasitic; with parasitics the group is -43.438 (\$7.2), so +4.925 is about **11% of the real violation**. **The precondition gate fired for the first time**: A6 declared k = 0 while splitting a 15-entry array in two, the flop count moved by +256, and G3 rejected it before any solver ran. And **A5 is why we registered a control**: it unrolls a reset loop, touches nothing on the read path, and still moves clk\_b by +0.436, so A4's honest figure is **+4.489 above a change that does nothing**.

**A2 is the batch's real finding, and it is a bug in our gate**, told in §9: EQY prints the same "Failed to prove equivalence" line for a counterexample and for a depth bound, and we matched on the string. Three things did not fit, and the one that mattered was that EQY had proved **128 of 128** partitions feeding the output it failed. A3's REFUTED was re-checked under §9's null control and holds (§7.6).

## 7.2 The second router: which lever, before which transform

Four of batch 1's six proposals were proven correct and three made timing *worse*. The obvious reading is that LLM RTL proposals do not help. The measured reading is more useful.

tools/classify\_path.py scores what fraction of a path's delay comes from cells driving 32 or more loads. Batch 1's target path is **58.9% fanout-attributable**; the design-level AES path is **91.4%**, with 21.029 ns in a single nor4\_1 driving **300 loads**. Those proposals were aimed at the wrong variable, though not at an untouchable one: §7.6 and §7.7 show two proven RTL transforms moving that same 91.4% path.

The control was registered in advance: synthesize bench\_top twice from identical RTL under identical SDC, differing only by appending buffer -N 16; upsize; dnsz to the ABC script, which Yosys ships in its -liberty -constr script and our flow was not running. Both violated groups close, clk\_b **-4.957 to +12.600 MET**, zero RTL change, 1,419 buffers, identical flop count. Checked rather than trusted: **49,923 of 49,924** obligations discharged, the residual a top-level XOR undriven in *both* designs.

That control has no parasitics. With them, **all three groups close** and the closure survives a clock tree and global routing; the numbers, the area cost and the required-period table are in §8, which is where deliverable 5 lives.

The *before* column carries the project's largest correction (§9): every timing number published before this experiment was **zero-parasitic**. That strengthens the routing argument rather than weakening it, because wire delay is definitionally not an RTL problem.

On one group under one SDC the ladder runs: best batch-1 RTL transform **+0.485**, best batch-2 **+4.925**, both zero-parasitic; ABC's buffering control **+17.557**; and **repair\_design +55.805**, the only one with parasitics. The mapping-level control pointed the right way and understated the real pass by **3.2x**, which is what a control is for.

Two results keep this from being a simple "buffering wins" story. **A4 gained 4.925 ns while leaving max fanout exactly unchanged** (2,193 in gold and in every proven variant), as pre-registered in H2, because synthesis re-merged its duplicated cones. **That re-merge is not universal**: the unattended proposal in §7.7 duplicates a control cone and the mapper keeps it, for +1.414 ns. And **after buffering the remaining violations are mixed, not depth-dominated**,

first reported as DEPTH at 0.0% and corrected (§9) to **34.4% and 42.8%** on the tightened fixtures and **42.8% and 28.6%** in the v3 loop. **The two levers are sequential, not alternative.**

**Does any of this transfer off our own benchmark?** Pre-registered on the 20 human-written designs published with Dr. RTL, same flow, same thresholds, each at 0.9x its own measured requirement. 15 are in scope and split **5 FANOUT / 3 MIXED / 7 DEPTH**, byte-identical across two executions. The physical lever **closes all 5 fanout-dominated designs alone**, median **3.623 ns**, against 4 of 7 depth-dominated at a median of 0.581; buffering alone does **nothing** for depth-dominated designs (median 0.000, worse on 3 of 7), which is the claim the classifier actually makes. **The primary prediction was wrong**, because upsize; dsize is gate *sizing* and sizing helps any path (per-prediction verdicts: tools/tally\_predictions.py). Applying Dr. RTL's own two high-confidence fanout skills as written, **0 of 4 applications both reduced the worst path's fanout and improved timing** (experiments/drctl\_transfer/).

**Does the router ever fire on its own?** Not on this benchmark, and that is a property of the benchmark rather than of the router. So we pointed it at i2c\_master\_top from the Dr. RTL set, 560 cells, previously classified **DEPTH** at fanout share 0.000, with the target frozen at 0.9x its own measured requirement before the loop ran (experiments/unforced/, registered first).

```
classify wb_clk_i: DEPTH_DOMINATED (fanout share 0.0) -> rtl
```

**No --force-lever, and no human at any step.** The complete run reads: measure -0.396, classify DEPTH\_DOMINATED, route to RTL, propose, gate, **refuse**. The gate reported UNRESOLVED and the loop declined, though §7.8 shows that verdict was almost certainly a gate defect; its artifacts were not kept. Three runs returned *three different* transforms, all aimed at the wide one-hot case compare the classifier reported; a hand-run EQY proof of one left no artifact and is not counted. experiments/depth\_i2c/ (§7.8) re-runs it with everything kept.

**So the router fires and the engine proposes; it has not yet improved this design.** Same model, prompt and design gave three different transforms, so "one sample per proposal" (§10) is a measured limit, not a disclaimer.

**Getting there took six runs, and every failure was a defect in our own tooling** that bench\_top structurally could not expose: the classifier scored the **first** path block rather than the worst; the loop resolved a module by **filename** and the gate suffixed only the target module (our benchmark is one module per file, i2c.v has three); the prompt showed the proposer a *different path* than the router acted on; target\_file carried a hardcoded rtl/ prefix; and the CLI timeout was short for a 25 KB module. None touches a published result. **Two were turning a proposal EQY proves into a reported harness error.**

That cuts both ways and this report should say so: it is evidence that the router works, and that the tooling had never left its own benchmark.

### 7.3 The closed loop

tools/slacksmith.py runs all of the above as one command (§1.1), logging every decision with its evidence to decisions.jsonl.

Against SDC v2 it closes in **2 iterations** (wall clock in §1.1), and the RTL lever never fires because nothing is left for it. To exercise both branches the target must be one the flow cannot already clear, so sdc/bench\_top\_v3.sdc applies v2's methodology to the corrected flow (sdc/make\_v3.py). Under v3 the physical lever alone closes clk\_b outright, -18.957 to +5.6; the RTL lever then gates P1, P2 and P3, EQY proves all three, and **the loop reverts all three on G5**. The default -lever-policy blunt applies buffer; upsize; dsize as one step although buffer alone meets more groups (§8); --lever-policy verdict, used by run\_v3\_fixed, lets G5 revert the sizing half. The default is unchanged.

**Rankings change once buffering has run.** P2, batch 1's only clk\_a improvement at +0.485 ns, was reverted. Isolated on one SDC and one transform differing only in whether buffering runs: unbuffered **+0.485**, buffered **-0.485**, sign flipped. **Batch 1's single winner is a loser in the context the design would ship in.** One transform, one setting; the equal magnitude is reported, not claimed as a law.

**The acceptance bar itself was wrong.** A per-group G5 bar confirmed a sizing step gaining 0.838 ns on clk\_e while costing clk\_a 3.466; measured across all groups it is reverted, **4 iterations instead of 8, 1 group violating instead of 2**. Both runs are kept.

**Under v3 with a corrected classifier the RTL lever never fires at all**, so the runs above measured three proven transforms on a path the router should not have sent them to. The flow was the larger lever: **flattening alone moves clk\_a by +22.446 ns**, because across the module boundary ABC collapses decode logic our wrapper's tied instruction bits make redundant (experiments/flatten\_control/).

**repair\_design's output is formally proven equivalent to its input:** 5,832 compare points, all proven, **38 seconds**. Four earlier attempts failed and were published at the time as an open limitation; every cause was mundane (equiv\_make matches wire names, so hierarchical against flat gave 86 compare points; a k-padded miter asked a sequential question of a combinational change). This is translation validation per run, not a proof of the algorithm, and says nothing about whether the timing gain is real.

**Running the loop found three defects in it:** it gated a module not on the binding path and kept a -0.487 ns result for want of a G5 bar; a string-match fix then read P4's counterexample as UNRESOLVED. The standing regression: **P4 must read REFUTED and A2 UNRESOLVED**. Pre-fix logs are kept.

**The proposer was offline until 5 Sept because we conflated two things:** pre-registration forbids the experimenter changing the hypothesis after seeing results; it does not forbid the system proposing in response to a measurement, which is the loop working.

**Two online proposals, both proven, one kept and one rejected by measurement.** O1 moved `clk_e` -25.957 to -24.079 and was confirmed; O2 was **also PROVEN and made the design 11.434 ns worse**, reverted on G5. Proof and profit are independent questions, shown inside one run on a transform written minutes earlier. "Online beats frozen" is reported **inconclusive at N = 2**.

**What it does not show.** The router never chose RTL here; `--force-lever rtl` overrode it every iteration, logged as `lever_forced`. O1's stated mechanism was wrong while its number was real. And revert restored *pristine* source rather than the last confirmed variant, silently discarding O1's +1.878 ns, a defect no frozen run could reach.

## 7.4 SlackBench: we built the exam and published our own score

Every RTL benchmark we found grades a *design* or a *testbench*. **SlackBench grades a verification methodology:** eight transform pairs, ground truth and trap class committed **before any checker ran**, each built to defeat a specific checker's abstraction. The score is a confusion matrix, never one number, because a checker that rejects everything would otherwise win. Totals over 8 cases, case-by-case in `NOTES.md`:

| checker                                 | correct | wrongly ACCEPTED | wrongly REJECTED | cannot express |
|-----------------------------------------|---------|------------------|------------------|----------------|
| cec / dsec                              | 2       | 0                | 1                | 5              |
| EQY                                     | 3       | 0                | 0                | 5              |
| sim lazy / aggressive                   | 6 / 7   | 2 / 1            | 0                | 0              |
| miter, induction / <b>PDR (shipped)</b> | 6 / 7   | 0                | 2 / 0            | 0 / 1          |

Four findings. **A wrong transform survived 40,000 simulated cycles**, wrong on one input pair in 65,536, with that pair published in advance. **Combinational and sequential EC could not express 5 of 8 questions**, each refusal evidenced by latch counts. **cec and dsec both confidently rejected an equivalent pair**, because they match latches positionally: a checker answering a different question than the one asked can be wrong without signalling it changed the question. And **a k-padded miter is wrong twice under temporal induction and zero times under PDR**, because induction quantifies over states no execution reaches. The shipped gate discharges with BMC plus PDR and accepts only on PDR, scoring **7 of 8 with one decline**; the induction row is kept published rather than dropped once it looked worse. **The engine that never lies is the one that sometimes refuses.**

**One of six registered predictions is wrong:** EQY declines both STIMULUS cases rather than refuting one. Prediction 6 registered that our own gate should not sweep its own exam, and it did not. An X-propagation addendum adds one more: on identical stimulus a testbench comparing with `==` accepts a dropped reset while `!=='` catches it in 50 cycles, so **the verdict is a property of the comparison operator**. `experiments/slackbench/`.

## 7.5 G7: the gate for the defect equivalence checking cannot express

§7.4 leaves both CDC cases unsolved: every checker there is either unable to express the question or **correct and useless**, since CDC-2's pair really is functionally equivalent and the bug is still there. So we built a seventh gate (`experiments/cdc_gate/`, registered before the code).

G7 checks two things on elaborated RTL. **Synchronizer depth**, structurally: fewer than two back-to-back flops in the destination domain is a missing metastability guard. **Hamming safety**, temporally, via SymbiYosys: the value crossing a boundary must change at most one bit per cycle. The second checks the **property, not the encoding**, which is what makes it work: CDC-2's gold crossing net is a combinational wire and passes because its value changes one bit at a time, not because anything pattern-matched a gray encoder.

| case                                                                                   | every checker in §7.4                               | G7                       |
|----------------------------------------------------------------------------------------|-----------------------------------------------------|--------------------------|
| CDC-1 gold                                                                             | cec/dsec cannot build a miter                       | SAFE, depth 2            |
| CDC-1 gate                                                                             | reads as a latency change, not a defect             | <b>DEPTH_1</b>           |
| CDC-2 gold                                                                             | correctly ACCEPT                                    | SAFE, proven to depth 16 |
| CDC-2 gate                                                                             | correctly ACCEPT, <b>and the bug is still there</b> | <b>REFUTED</b>           |
| The counterexample is not about function: the crossing bus goes 0001 to 0010,          |                                                     |                          |
| <b>two bits in one cycle</b> , so a receiver in another domain sampling mid-transition |                                                     |                          |
| can latch 0000 or 0011, neither the old value nor the new one.                         |                                                     |                          |

**Six registered predictions, three confirmed and one plainly wrong.** We predicted zero depth violations on `bench_top` and got **six**, every one a clock crossing to its own in-RTL divided version. Those are synchronously related and the design is correct. The scorecard first recorded **"6 of 16 findings are noise" and the true figure is 8**: both UNCLASSIFIED entries were also synchronous, so the earlier number counted false *violations* and missed two spurious non-verdicts.

**The cause was that G7 keyed on clock *nets* with no notion of clock *relationships*, and the fix was already in the constraints:** `create_generated_clock -source` declares every derivation in the file **Go fingerprints**, so `--sdc` reads clock groups from constraints Go has vouched for. A crossing inside a group gets a SYNCHRONOUS verdict and is excluded:

| verdict                                | without --sdc | with --sdc |
|----------------------------------------|---------------|------------|
| DEPTH_1, all false                     | <b>6</b>      | <b>0</b>   |
| UNCLASSIFIED                           | 2             | <b>0</b>   |
| SYNCHRONOUS                            | n/a           | <b>8</b>   |
| MULTIBIT, the real FIFO gray pointers  | 6             | 6          |
| SAFE, the real async control crossings | 2             | 2          |

**All eight spurious findings go away and no real one does.**

**Then the exit code was lying.** With clock groups the run exited **0** while six multi-bit crossings sat undischarged; MULTIBIT means not checked. Both it and UNCLASSIFIED now exit non-zero naming what was skipped, `bench_top` exits **1**, and its six gray pointers are discharged on `async_fifo` itself, **both PROVEN to depth 16. Three of G7's six bugs produced a confident wrong verdict rather than an error**, caught only because gold ran through every check beside gate. **A checker exercised only on the case expected to fail is indistinguishable from one that always fails.**

## 7.6 Two of the four named classes were unreachable by construction

The objectives name pipelining, logic restructuring, **retiming** and **FSM optimization**. For most of this project the engine proposed the first two, and the explanation on the record was that nobody had written the others. **That explanation was never tested and it was wrong.**

G3 required  $k = 0 \Rightarrow \text{flop delta} = 0$ , with no exception. A retiming moves a register across combinational logic:  $k = 0$ , flop count changes. A state re-encoding widens a register:  $k = 0$ , flop count changes. **Neither could pass whatever its content.** Worse, `proposer_prompt.md` *advertised* branch 4 to the model while every  $k = 0$  proposal was routed to EQY before its declared branch was read, so a proposer that followed the template was guaranteed a rejection. Measured on our own published one-hot domain\_b, RTL unchanged: FAIL(declared  $k=0$  but flop count changed by +12) through the old gate, PROVEN in 64 s through the fixed one. That is why `experiments/fsm_reencode/` carries a hand-written miter and never invokes the gate.

Branches 4 and 5 are now implemented, both  $k = 0$  with the flop delta unconstrained, both discharged by the **sequential miter** rather than EQY, because neither leaves a flop correspondence for EQY to pair internal nets across. Against `aes_key_mem`, the module the loop binds on `clk_b`:

| proposal                                       | class            | branch | verdict             | clk_b                                                                         |
|------------------------------------------------|------------------|--------|---------------------|-------------------------------------------------------------------------------|
| <code>retime_write_decode_forward</code>       | retiming         | 5      | PROVEN (PDR, 222 s) | <b>-4.616</b> (and -4.616 on <code>clk_e</code> , +0.274 <code>clk_a</code> ) |
| <code>fsm_output_coded_state_assignment</code> | FSM optimization | 4      | PROVEN (PDR, 124 s) | <b>+3.185</b> (and +3.185 <code>clk_e</code> , +0.528 <code>clk_a</code> )    |

**Both read "PROVEN, 2 of 3 outputs" for a day, and the partiality was ours.** The miter gave two instances *independent* arbitrary power-up state, which asks whether they agree from any **pair** of starting states: not equivalence, and unsatisfiable by any correct transform. `setundef -init -zero` fixes it: gold against **itself** went from failing `eq_round_key` in 1 s to passing, and both proposals became **unbounded PDR proofs over all three outputs**. The assumption is zero, not "the same arbitrary value" (Yosys rejects a hierarchical assume here), and it is written into every result JSON.

**What makes those two proofs mean anything is the check on a transform known to be broken.** A3 stays REFUTED under the same assumption, failing `eq_ready` in 1 s. An assumption that proved a known-bad transform would prove anything, and that void condition was registered before the assumption was written.

**The retiming is the worst RTL transform in the project and the FSM re-encoding is the best.** R5 predicted the retiming would not improve its group; it does not merely fail, it costs 4.616 ns. R18 predicted the FSM transform would not materially improve `clk_b`; it bought **+3.185 ns**, the largest RTL gain here, and it is the measurement that falsified §1's absolute.

**Getting there took three tries** and is why the null control in §9 exists. It also cleared a confound: A3 (§7.1) fails on `eq_ready`, an output the control **proves**, so it was never the artifact. **P5 took six attempts.** Its control does not close on `rv32i_core` at any budget tried (depth 20 and 5, 300 s and 1800 s, both instances zeroed): on two copies of 2,048 flops this harness **finds** a counterexample in 1 s and cannot **prove the absence** of one in 1800. But the one artifact class found in our own miter, differing power-up states, is ruled out for P5: remove that freedom and the refutation survives. So **P5 is REFUTED with a named limit.**

## 7.7 The unattended run, N = 1

Until 2026-09-11 the `cli` backend was committed and never executed; the blocker was an expired credential, now held outside the repository, with `tools/preflight.sh` failing if a credential-shaped string reaches a tracked file. **The first**

**attempt died before the model was reached** on an empty binary path from a non-login shell; registered as an environmental failure, logged, retrieved.

**The second attempt ran the loop end to end with no human in it**, 456 s: measure clk\_b = -18.957, classify FANOUT\_DOMINATED at 0.9139, propose, gate G4=PROVEN, G7 skip, apply. **Three qualifiers belong in this sentence and not in a footnote**: the lever was --force-lever rtl, a human flag, because the classifier correctly routes this path to physical; the run was capped at --max-iters 1, so the loop's own accept-or-revert step never ran on the proposal; and the +1.414 ns below was measured afterwards by time\_01.sh, outside the loop. The unforced routing claim is \$7.2's, on a different design. Request, raw response, variant and decision log are at experiments/cli\_backend/results/run1/.

Four predictions were registered before the run. **Three held and the fourth was the interesting one**. C1, valid JSON first attempt, held despite two warnings and a failing hook message mixed into the captured text. C2, a G4 verdict with no human, held. C4, logic restructuring rather than retiming or FSM, held.

**C3 said the proposal would not materially improve clk\_b, and gave a mechanism in advance: the duplicated control nets are aliases and opt\_clean merges equivalent nets. Both were wrong.** The model's own rationale identified that round\_key\_update gates roughly 384 bit-positions and matched it to the 300-fanout nor4 the classifier had just reported, then split the cone so the two wide select networks are driven separately. The cone duplication is not an alias and the mapper keeps it: **+1.414 ns on clk\_b, +1.414 on clk\_e, +1.967 on clk\_a, no group paying for it**, under a proof EQY discharges over all outputs.

That is the only batch-3 transform that improved every group, and it has the strongest proof of the three (\$7.6). It is also **N = 1**: one design, one sample of a nondeterministic generator, one run. The claim is that the automation path works, not that it works reliably.

**7.8 The composed RTL, what survives the physical lever, and the depth-side control**

Deliverable D4 now ships one optimized file, aes\_key\_mem\_composed.v: A4, O2 and O1 merged three-way against the gold file by compose.sh, rebuild-checked, **PROVEN** on branch 4 by PDR under \$7.6's zero-init assumption (experiments/composed\_rtl/, registered before synthesis). Measured four ways on clk\_b, one SDC, one liberty, null control 0.000:

| composed minus gold, clk_b                           |                                                     |
|------------------------------------------------------|-----------------------------------------------------|
| zero-parasitic, unbuffered                           | +5.165 (arithmetic sum of the parts +9.524, so 54%) |
| after the ABC buffering lever (buffer; upsize; dnsz) | 0.000                                               |
| with placement parasitics, unbuffered                | +18.792                                             |
| after repair_design                                  | -0.237, against a perturbation floor of 0.24        |

The floor is measured, not assumed. Gold through the physical flow twice is identical to every digit; five perturbed netlists (a do-nothing edit, each transform alone, the composition) span **0.241 ns** on clk\_b and 2.4 ns on clk\_a. Every transform-bearing netlist lands at most 0.004 ns above gold after repair (A4) and the do-nothing edit lands at -0.006, so the direction, where it resolves at all, is not positive. R38, "the marginal gain is positive", was **WRONG**, then **VOID** under a registered amendment because it sits inside the floor; one registration in that amendment was itself flawed and is scored as written with the flaw disclosed. Area after repair moves both ways across the five (-5,167 to +1,799 u2), so no area claim is made either. **Stacking proven transforms does not stack their gains, and on these paths nothing stacks with buffering at all.**

**The classifier is a cost-saver, not a decision procedure: on both depth-dominated designs it routed to RTL, and no proven transform helped.** That was the fanout-dominated row of a two-row table. The depth-dominated row, the same measurement on the one design the classifier routes to RTL unforced, is experiments/depth\_i2c/ (R54 to R60, registered before running): three unattended runs, no forced lever, all routed to RTL by the classifier. Run 2 **proposed, proved, applied and reverted** a (\* parallel\_case \*) transform that synthesizes to a **byte-identical netlist**, 0.000 ns in every column; every decision correct, gain zero. But a do-nothing edit moves this 560-cell design by 0.424 ns, so it could not have shown a gain. **So we repeated it on tv80s** (experiments/depth\_tv80/, 3,447 cells, floor **0.152 ns** measured before predicting, physical lever does not close it): routed RTL unforced **3 of 3, 2 of 3 PROVEN**, and **both reverted**, one byte-identical to gold (a parallel\_case hint again, which synthesis already infers) and one **-0.268 ns**. **The depth cell is empty on the design that had room to fill it. Both designs' run 1, the only proposals aimed at the path's depth, were never gated**: the gate built the miter on our RV32I port list and reported the error as UNRESOLVED (experiments/invariant\_obligation/; three gate defects, no PROVEN verdict changed). Re-gated, the tv80 proposal holds only if MCycle[5] and MCycle[6] are never both set. We proved that on tv80\_core by k-induction and assumed it at the child: **PROVEN**, a broken variant still refuted, the sole unassumed counterexample exactly the unreachable input, and **0.421 ns worse**. Four proven transforms: 0.000, 0.000, -0.268, -0.421.

**An open-weight model through the same loop** (experiments/open\_weight/, registered first): Qwen2.5-7B-Instruct under Ollama, no key, no vendor, the identical request. 1,172 s of CPU inference returned invalid JSON, part VHDL, interface not preserved; it never reached G1. **Qwen2.5-Coder-7B** failed identically (1,335 s, invalid JSON); with Ollama's JSON mode, registered in advance as the follow-up, it returned **valid JSON in 160 s and failed G1**. Across two models and three runs no proposal reached the formal gate, **but every run was cut short**: Ollama's default context evaluated **2,050 of the request's 5,113 tokens**, so each model saw 40% of the task (experiments/open\_weight\_3/). Our reading that the wall was the engineering, not the envelope, is withdrawn. Replayed with the whole request it

answered in 3,025 s, past the loop's 1,800 s wait, and a hosted 31B attempt was rate-limited: both void, and a hosted run is still to do. Nothing here speaks for larger open-weight models, and the prompt was tuned against Claude Opus 5.

## 8. Optimized RTL, frequency and PPA

**Design level, with placement parasitics.** Required period comes from the **capture clock** named in each path report, not the group's create\_clock period: after repair\_design the clk\_b group is captured by **clk\_b\_div3 at 79.5 ns**, not clk\_b at 26.5, so reading the group name would have reported clk\_b three times faster than it is.

| group        | capture clock     | period | slack          | required period | F_max             |
|--------------|-------------------|--------|----------------|-----------------|-------------------|
| clk_a before | clk_a             | 30.000 | -36.723        | 66.723          | 14.99 MHz         |
| clk_b before | clk_b             | 26.500 | -43.438        | 69.938          | 14.30 MHz         |
| clk_e before | clk_e             | 26.500 | -47.683        | 74.183          | 13.48 MHz         |
| clk_a after  | clk_a             | 30.000 | <b>+17.593</b> | <b>12.407</b>   | <b>80.60 MHz</b>  |
| clk_b after  | <b>clk_b_div3</b> | 79.500 | <b>+12.367</b> | <b>67.133</b>   | <b>14.90 MHz</b>  |
| clk_e after  | clk_e             | 26.500 | <b>+19.529</b> | <b>6.971</b>    | <b>143.45 MHz</b> |

Five asynchronous domains have no single F\_max, so the design-level figure is the factor **k** by which every period must be scaled for all of them to meet:  $k = \max(\text{required} / \text{period})$ . Before, **k = 2.799**, binding on clk\_e, so the design runs at **0.357x** its SDC target. After, **k = 0.844**, binding on clk\_b, so it runs at **1.184x** target with margin. **The flow improves achievable frequency by 3.32x. Read that against §9:** the *before* column is a flow that was not running the buffer; upsize; dsize script Yosys ships in -liberty -constr, worth 17.557 ns on its own. A meaningful part of the 3.32x is a flow defect we shipped, not a lever we invented, and the same is true of the +55.805 ns closure.

Two honesty notes. The binding domain **moves** from clk\_e to clk\_b, so before-and-after F\_max for any single group is not a like-for-like comparison; the scaling factor is. And only clk\_a, clk\_b and clk\_e are reported, because clk\_c and clk\_d met at baseline and were never in the optimization loop.

**And the closure survives a clock tree.** Those numbers use ideal clocks, so we ran CTS and global routing on the same flow:

clk\_a / clk\_b / clk\_e post-place with ideal clocks: **+17.593 / +12.367 / +19.529**. Post-CTS with propagated clocks: **+17.616 / +8.694 / +18.428**. Post-global-route: **+17.117 / +8.987 / +18.647**.

Every group meets at every point, for 7,833  $\mu\text{m}^2$  (+1.45%) of clock tree and 1,547 clock buffers. clk\_b pays 3.673 ns for its tree, matching its **3.514 ns** launch-capture depth imbalance to 0.16 ns. report\_clock\_skew printed empty on this build, so no skew number is claimed.

**Core level, and the rankings invert.** On rv32i\_core alone P2 is smallest and lowest-power (6,441 cells, 6.20 mW against 6,769 and 6.26) and P6 has the best core timing; **at design level P6 is the worst** (-1.615 ns) and P2 the only winner, because the core's critical path is not the design's. Earlier drafts quoted a core-level **F\_max gain of 11.0% for P6**: the most favourable framing available for the least useful result, replaced by the table above.

| metric      | before                  | after                   | delta         |
|-------------|-------------------------|-------------------------|---------------|
| clk_a WNS   | -36.723 ns              | <b>+17.593</b>          | +54.316       |
| clk_b WNS   | -43.438 ns              | <b>+12.367</b>          | +55.805       |
| clk_e WNS   | -47.683 ns              | <b>+19.529</b>          | +67.212       |
| total power | 94.5 mW                 | <b>139.0 mW</b>         | <b>+47.1%</b> |
| design area | 448,840 $\mu\text{m}^2$ | 539,351 $\mu\text{m}^2$ | <b>+20.2%</b> |

**Closure costs 47% more power and 20% more area.** The expectation was registered before the run: 960 added buffers should raise internal and switching power, and both rose (experiments/ppa/power/ carries the derivation). Vector-free at default activity: it compares two netlists under one model and is not a signoff number.

**That +20.2% is one point on a curve, and not the best point.** Three repair arms on the same placed netlist, registered before running (experiments/closure\_cost/; SDC v3, so these slacks are not the v2 numbers above, while the areas are the same fixture):

repair\_timing -setup alone reaches a worst group of **-0.947 ns at +6.90% area and 0.313 W**; repair\_design, the row above, **-1.471 at +20.17% and 0.408 W**; both in sequence **-0.777 at +21.26% and 0.414 W**. repair\_timing after repair\_design **halves the remaining violation for +0.91% area**; targeted repair alone holds every group inside **0.947 ns for a third of the area**, meeting none outright. **No arm meets all three**, and on worst-group slack the published flow is last. Zero-parasitic, the same experiment found the loop's ABC lever is not its best half: buffer -N 16 alone meets **2 of 3** groups where buffer; upsize; dsize meets **1 of 3**, trading a met clk\_a for 3.6% area; sizing alone meets clk\_b for **+1.39% area** and no added cell. Seven predictions, four confirmed, three missed.

At the RTL-variant level power is a **null**: flat at 223 to 224 mW across all four proven transforms, because a 351-cell change is 0.6% of a 55K design. Both facts belong in the same section: the transforms we proposed do not move power, and the pass that closes timing moves it by half again.

## 9. What we got wrong

**Our gate manufactured a refutation three times, so we stopped fixing them one at a time.** It reported A2 as REFUTED when EQY had only run out of depth (one partition failed while all 128 feeding it passed). It compared two independently uninitialised memories and read that as non-equivalence (experiments/g7\_in\_loop/). And the sequential miter was found **refuting aes\_key\_mem against itself**: Yosys does not apply a module's async reset to a memory, so two instances start from different arbitrary contents; gold versus gold, FAIL eq\_round\_key in 1 s.

Three occurrences is a process defect, so the fix is structural: **a null control now runs before any refutation is reported**, the same miter with gate replaced by gold, and if that also fails the gate reports CANNOT. **Its first three versions were also wrong**, two caught by registered predictions (R10, R13). Four iterations on one control.

**We defended a wrong explanation for a day (§7.6)**: two proofs reported partial, and **three written explanations blamed the design before the harness was found at fault**, the direction that flatters the tool. No registered prediction caught it.

**Registered predictions we got wrong: 35 of the 107 decided, across 136 registered**, and 8 not scored. The tool reads lettered ids, which appear in 15 of the 24 registered experiments; SlackBench, the transfer study and batch 1 are outside these counts. **The counts come from tools/tally\_predictions.py, and until 13 Sept the tool itself was wrong**: it counted every mention of an id (38 of 102 across 153), and until 14 Sept it missed the word CORRECT; it now self-tests on every line that broke it. Before that, counts went stale twice, and check\_report\_numbers.py passed both, because it asks whether digits appear, not whether a tally is current. **The page count failed the same way twice**: estimated from pixel height, then measured on a stale render of this file; tools/page\_count.py now renders the report and prints a real PDF every time.

**We spent most of the project optimizing a design whose violations a stock pass closes, and reporting numbers with no wires in them.** Yosys ships buffer; upsize in its -liberty -constr ABC script, not the plain -liberty one we used; it is worth 17.557 ns. And every timing number before §7.2 was **zero-parasitic**: with parasites the clk\_a baseline reported as "+1.333 MET" is **-36.723**. Baseline-versus-variant comparisons survive, since both sides used one model; the absolute closure claims did not.

**The path classifier undercounted fanout across module boundaries, and we had written down the tell and shipped it anyway.** Its docstring named "a 6.762 ns delay on a cell at fanout 1" as the signature, and that number sat in every v3 log under a DEPTH\_DOMINATED verdict. The cell drives **387** loads: whole-bus and concatenated port connections were charged nothing. Every DEPTH verdict in runs 2 to 4 is MIXED, 1 of 15 external verdicts changed, and the fix is regression-checked on 5 fixtures against OpenSTA's own fanout column. The wrong logs are kept.

**Smaller ones, all surfaced by running rather than reading.** Two harness bugs appeared as UNRESOLVED (sby off PATH; {max(k,1)} written into a miter as literal Verilog). The acceptance bar confirmed a step that traded a met group for 0.838 ns on another. A fresh clone on 2026-09-12 scored **13 of 15** because demo\_check wiped fixed scratch paths under concurrent runs; per-run paths now pass **15 of 15**. And pipeline\_cut\_rigid(domain\_a) was called boundary-proven *and therefore* module-equivalent, but the consuming domain samples at half rate and simulation showed mac\_result diverging (002a vs 0031), so the claim is withdrawn and the refuting testbench committed.

## 10. Demo, reproduction, limits

The demo ends on a refutation that every simulation-gated tool would have passed. DEMO.md is the shot list and tools/demo\_check.sh runs every command in it, failing if any breaks; it holds **23 assertions**, all passing on 14 Sept (beat 5 was missing until 13 Sept). **demo/explorer.html** is the interactive deliverable: five runs steppable, each decision with the evidence it used, generated from the logs so it cannot drift from them. **Reproduction, checked rather than asserted.** Three clean clones passed **12 of 12, 12 of 12 and 15 of 15** (experiments/reproducibility/). A fourth, on 13 Sept, **failed 21 of 22** because nine proposal files pointed into our scratch directory; the gate now reads the byte-identical copy committed beside each and records variant\_file\_resolved, and the re-clone passed 22 of 22. Each experiment directory holds its sources, .sby/.eqy configuration and raw logs; proofs re-run with sby -f <config> pdr.

**Limits we would rather state than be asked.** **N = 17 proposals in three provenance tiers (§3): 12 frozen**, written against a timing report and committed before any check ran; **4 handoff**, written by a model through the session driving this project and therefore carrying its context; and **1 unattended**, claude -p with no human and no context beyond the prompt. Three target modules, one proposer model (**Claude Opus 5**, disclosed in every registration, default sampling, one sample per proposal, no best-of-n): outcomes, not rates. Unattended operation is **N = 1**. The physical flow reaches CTS and global routing, not signoff. The classifier's thresholds were chosen on our benchmark and one external verdict is flow-sensitive. The best run leaves clk\_e short with no ABC lever left. Equivalence is proven for repair\_design's netlist pairs but **not** for the ABC buffering lever, where the method times out. The two batch-3 proofs and the composition's are conditional on a stated initial-state assumption, and P5's refutation is **uncorroborated** because its null control does not close. The composition's post-repair timing sits inside a floor measured with five perturbations, so its sign is not claimed (§7.8). Of the transforms proven correct, **P1, P3, P6, online\_proposer O2 and missing\_classes O1 each made their own path group worse**, and P2 flips sign depending on whether buffering has run (§7.3). **Proof and profit are independent questions, and we measured both.**

*A4, 15 mm margins, 9.5 pt serif, rendered by tools/render\_report.py and page-counted by tools/page\_count.py.*